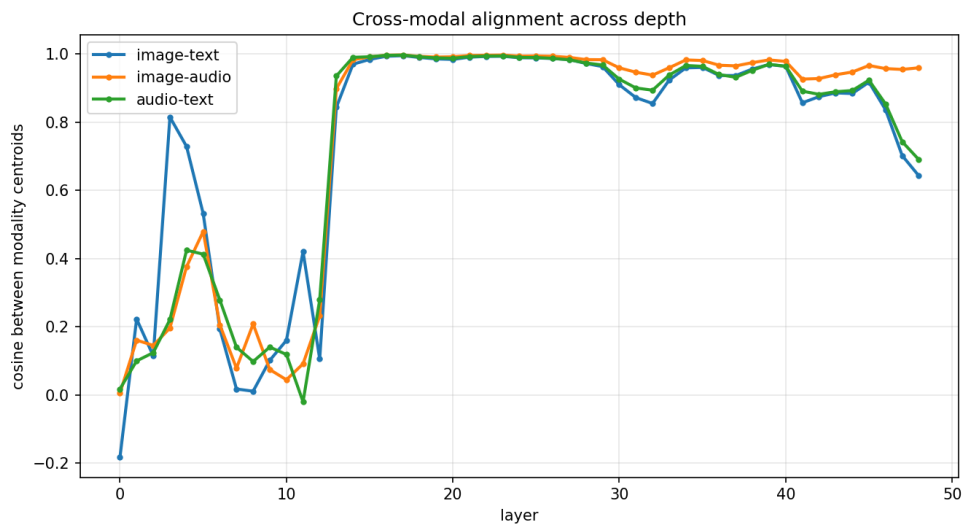


Gemma 4 12B

An Encoder-Free Multimodal Architecture — Findings Report

Local empirical study of **google/gemma-4-12B-it** — parameter anatomy, per-modality data flow, and representation-geometry analysis across all 48 decoder layers. All results reproduced on a single GPU from synthetic inputs.



Headline result: image/audio/text tokens begin in near-orthogonal subspaces and snap into a shared representation around layer 13.

1. Overview & a critical caveat

Gemma 4 12B (released 2026-06-03, Apache-2.0) is a dense, decoder-only multimodal model that ingests text, images, and audio **without dedicated vision or audio encoders**. Instead of a ViT/SigLIP tower or a Conformer, raw pixel patches and raw audio frames are projected directly into the language model's embedding space by tiny linear modules, and all modality “understanding” is performed by the shared transformer stack.

Critical caveat for researchers

The transformers library contains **two** Gemma 4 model types. `gemma4` (`Gemma4ForConditionalGeneration`) **has** a ViT vision encoder, a Conformer audio encoder, and MoE — it is the encoder-bearing sibling for other family members. The 12B-it checkpoint studied here is the separate `gemma4_unified` (`Gemma4UnifiedForConditionalGeneration`) implementation. “Encoder-free” is a property of this class and its config: the vision and audio configs contain **no** `num_hidden_layers` and no attention fields.

2. Parameter anatomy

Building the model on a meta device (config only, zero allocation) gives exact module shapes and parameter counts. The non-text paths are negligible in size — the empirical signature of an encoder-free design.

Component	Parameters	Share
language_model (decoder-only LLM)	11.91 B	99.56 %
embed_vision (entire image path)	49.9 M	0.42 %
- patch embedder (encoder replacement)	35.2 M	
- shared projection into LLM space	14.7 M	
embed_audio (entire audio path)	2.46 M	0.02 %
Total	11.96 B	100 %

Reconciling the “35M vision embedder”.

Google's materials cite a ~35M vision embedder; the full `embed_vision` module measures 49.9M. The difference is a module boundary, not a contradiction: the **patch embedder** (LayerNorm + Dense + factorized positional embedding) that actually *replaces the vision encoder* is 35.2M, while the remaining 14.7M is a generic RMSNorm → Linear(3840→3840) projection into LM space - the same module class the audio path uses (there as a 2.46M Linear(640→3840)). Thus: 49.9M (full path) = 35.2M (vision-specific) + 14.7M (shared projection).

Vision path — Gemma4UnifiedVisionEmbedder (no attention)

Images are patchified at 16 px, 3×3-pooled into 48×48 model-patches (6912 raw values each), then:
LayerNorm → Linear(6912→3840) → LayerNorm →
+factorized 2D pos-emb(1120, 2, 3840) → LayerNorm → RMSNorm →
Linear(3840→3840). Output: 280 soft tokens per image.

Audio path — Gemma4UnifiedMultimodalEmbedder (one Linear)

The feature extractor reshapes raw 16 kHz waveform into 640-sample (40 ms) frames. Each frame becomes one token via RMSNorm → Linear (640→3840). No convolution stack, no attention.

Decoder-only LLM

Property	Value
Layers / hidden / intermediate	48 / 3840 / 15360
Attention (GQA)	16 query heads / 8 KV heads
Hybrid pattern	40 sliding (window 1024) + 8 global, 5:1
RoPE	p-RoPE on global ($\theta=1e6$, partial 0.25); default on sliding ($\theta=1e4$)
Extras	QK-norm; K=V on global layers; 256K context; dense (no MoE)

3. Per-modality data flow (measured)

A single forward pass on a combined text+image+audio prompt produced the following input tensors and a unified sequence of 305 tokens (24 text + 256 image + 25 audio):

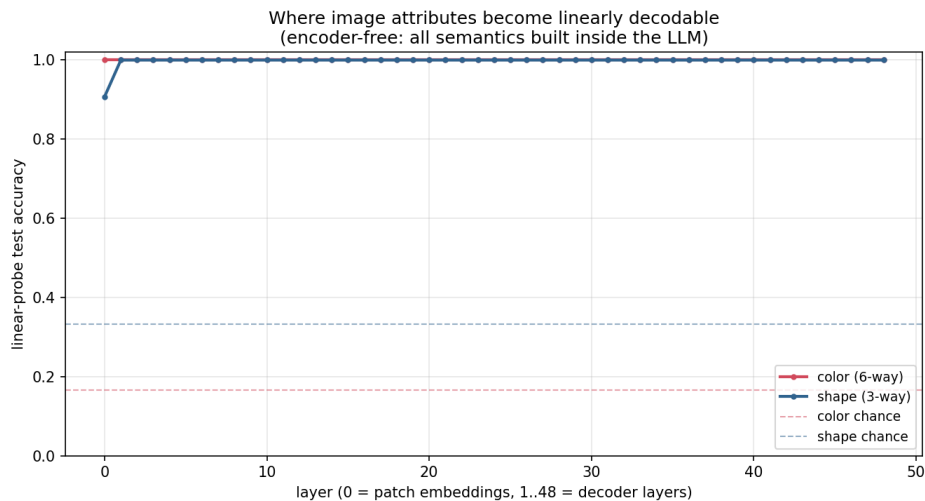
Tensor	Shape	Meaning
pixel_values	[1, 280, 6912]	280 raw 48x48x3 patches
image_position_ids	[1, 280, 2]	(x,y) patch coords
input_features	[1, 25, 640]	25 audio frames x 640 samples
input_ids	[1, 305]	merged token sequence

Both modalities are projected to the LLM width (3840) and scattered into placeholder positions, after which the same 48 layers process every token. Final logits: (1, 305, 262144).

4. Findings from representation analysis

4.1 Where image attributes become linearly decodable

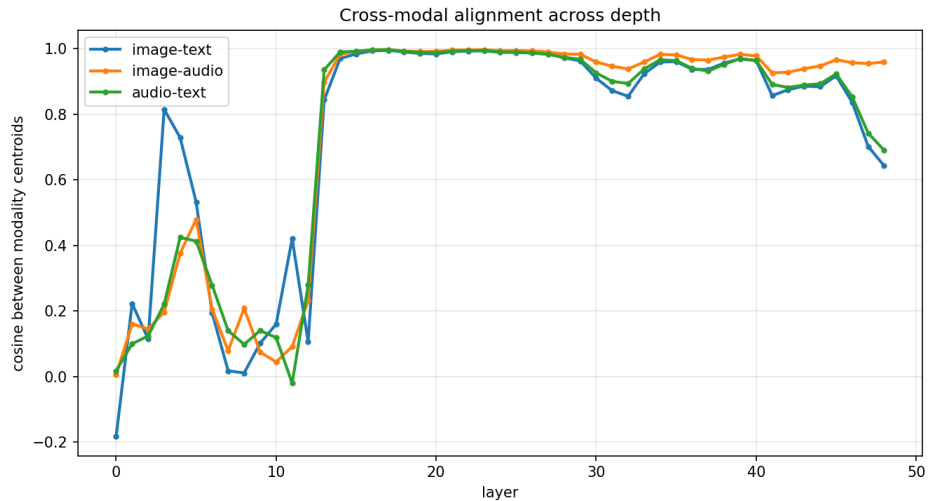
Linear probes on mean-pooled image soft tokens show **color is perfectly decodable at layer 0** (color acc = 1.00, the raw linear patch projection), while **shape** rises from 0.91 at layer 0 to ~1.00 within three layers. With no encoder, low-level appearance is linearly present at the embedder; spatial structure is assembled by the early decoder layers. The synthetic task is simple, so both saturate (ceiling effect) — for a graded curve, harder/natural attributes are needed.



Probe test-accuracy vs layer (color 6-way, shape 3-way).

4.2 Cross-modal alignment: a phase transition

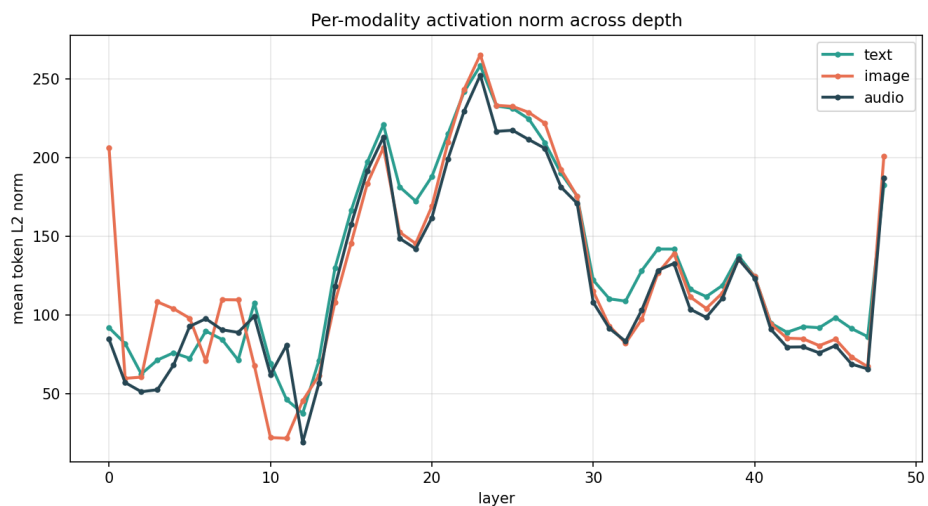
Cosine between modality centroids is near zero/negative in the first ~12 layers (image-text = -0.18 at layer 0), then jumps to ~0.99 around layer 13 and stays high through the mid-stack: the unified model **fuses modalities into a shared subspace**. Toward the output, image-text re-diverges while image-audio remains bound.



Cross-modal centroid cosine across depth.

4.3 Activation norm & the mid-stack bottleneck

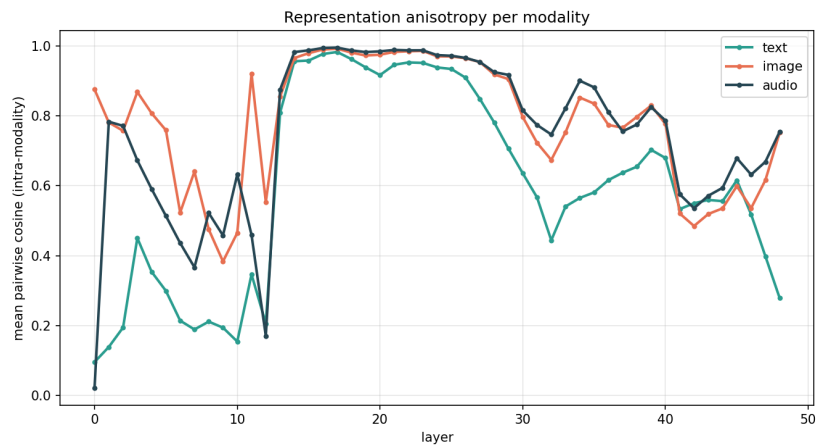
Per-modality token norms dip to a minimum near layer 11-12 (coinciding with the alignment pivot) and then peak sharply near layer 23 for all modalities — a massive-activation regime consistent with the per-layer absmx spikes recorded in the layer trace.



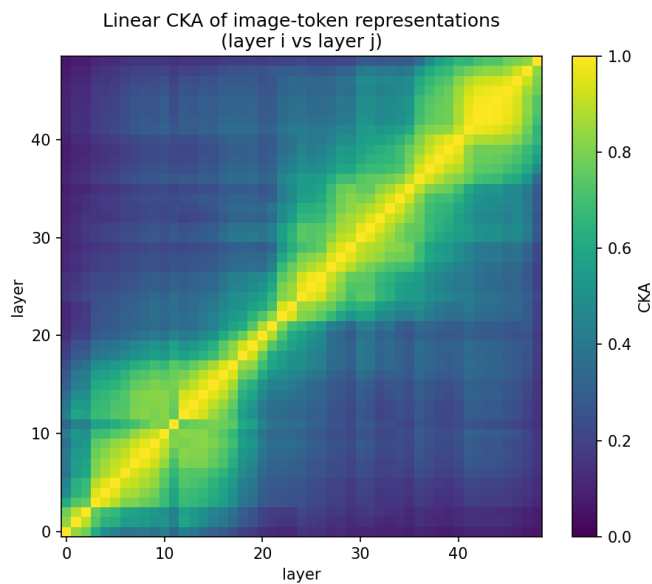
Mean token L2 norm per modality across depth.

4.4 Anisotropy & representational evolution

At the mid-stack peak, intra-modality mean pairwise cosine reaches ~ 0.94 - 0.97 (near-collapse to a shared direction), then partially recovers. Layer-vs-layer linear CKA of image tokens is high only locally (adjacent layers ~ 0.93) while $\text{CKA}(\text{L0}, \text{L48}) \sim 0.07$ — input and output representations are almost entirely distinct.



Intra-modality anisotropy (mean pairwise cosine).



Linear CKA of image-token reps, layer i vs layer j .

5. Research directions

- **Minimal modality adapters.** The whole vision “encoder” is 3 LayerNorms + 2 Linears + a position table (49.9M); audio is a single Linear (2.46M). This is an ideal test-bed for “what is the minimal viable modality adapter?” ablations.
- **The layer-13 fusion transition.** Characterize why modalities align abruptly mid-stack: causal interventions, per-head attribution, and whether the transition shifts with prompt structure or context length.
- **Massive activations at layer ~23.** Identify the outlier dimensions and their role (attention sinks / normalization), and whether they are shared across modalities.
- **Soft-token geometry.** Compare image (280) vs audio (40ms) vs text token manifolds: effective rank, norm growth, and per-modality re-specialization near the output.
- **p-RoPE + K=V global attention.** Analyze long multimodal-context attention patterns given the unusual proportional RoPE and key=value sharing on global layers.

Reproduce

scripts/01_architecture_report.py (static, no weights) · 02_modality_trace.py · 03_linear_probes.py · 04_representation_geometry.py · 05_build_report.py (this PDF). Environment pinned in requirements-lock.txt; weights from google/gemma-4-12B-it (ungated, Apache-2.0).